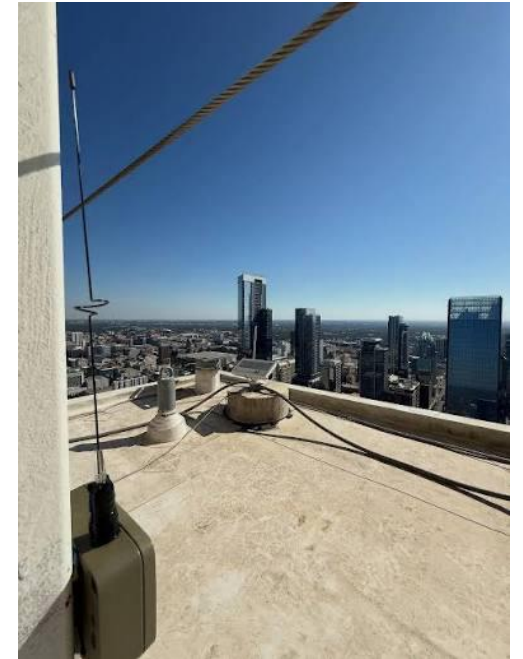
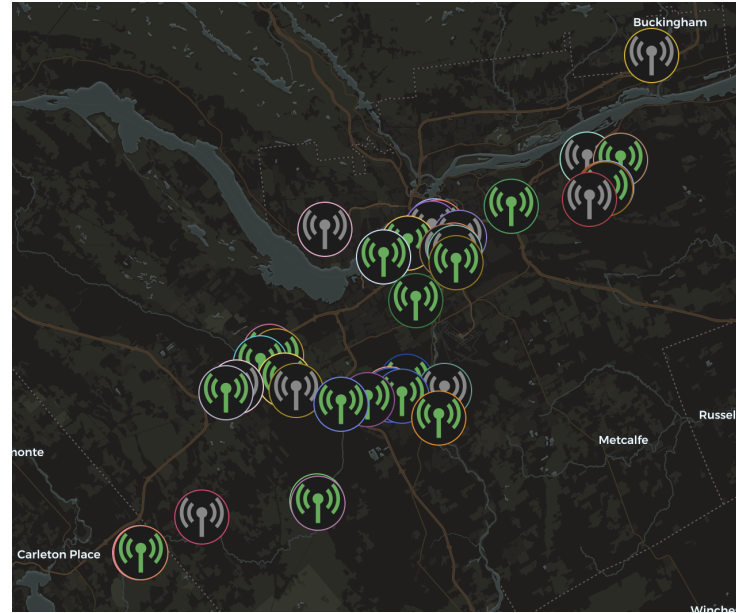


# LoRa Mesh Networking

Jon Earle

# Research Topic

- How does a real-world LoRa mesh perform in an urban environment compared to simulations?
- Which firmware and routing protocol thrive in such an environment?
- Can we realistically use LoRa meshes for more than just hobby?
  - Could we use it in an emergency?

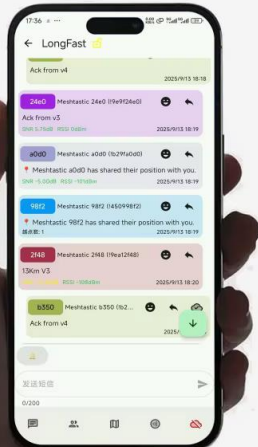


# Why LoRa Mesh?

- Accessible, easy to use, open-source firmware
  - Meshtastic, MeshCore
- Highly expensive, available firmware
- Strong, constantly growing community with plenty of supporting software
- Plenty of use cases, both recreational and emergency

## Connect to your Meshtastic App

<https://flasher.meshtastic.org/>



**Powered**

### Choose a Device

Meshtastic devices are available in a variety of configurations to suit your needs.

### Flash and Configure your Device

Flash your device with the latest version of Meshtastic and configure it to your liking.

### Connect to your Device

Connect to your device via any of our clients to start sending and receiving messages!



WiFi LoRa 32(V3), ESP32S3 ...

\$17.90 – \$19.90

Select options

WiFi LoRa 32(V4.3.1), ESP32...

\$17.90 – \$27.50

Select options



Mesh Node T114 (Rev. 2.0), n...

\$17.90 – \$33.70

Select options

Wireless Stick Lite(V3), ESP3...

\$14.90

Select options





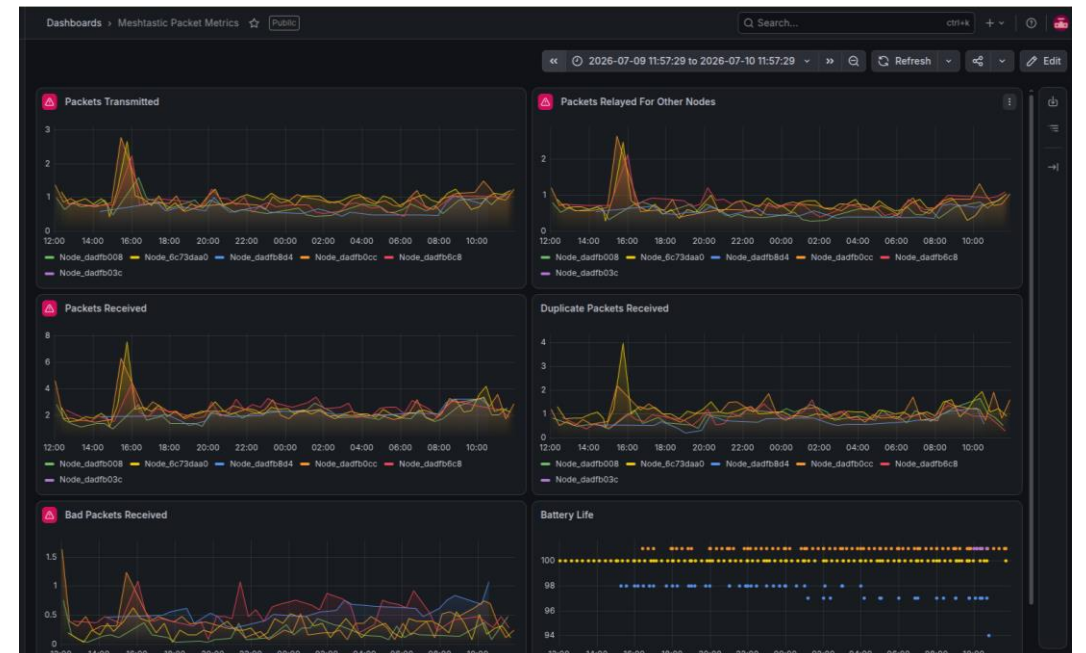
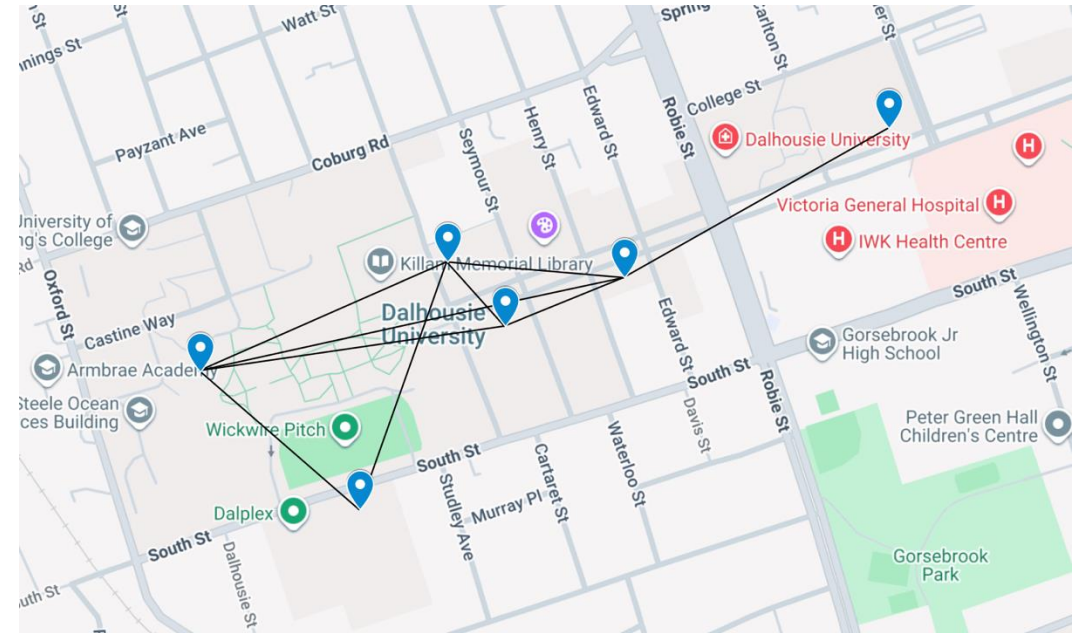
# A Challenging Problem Solved By LoRa Mesh

- Hurricanes and other natural disasters will become more common in the future.
- In the case of both a power and cellular outage, how will we communicate with loved ones?
- LoRa mesh is a low-power, inexpensive solution



# Dalhousie Deployment

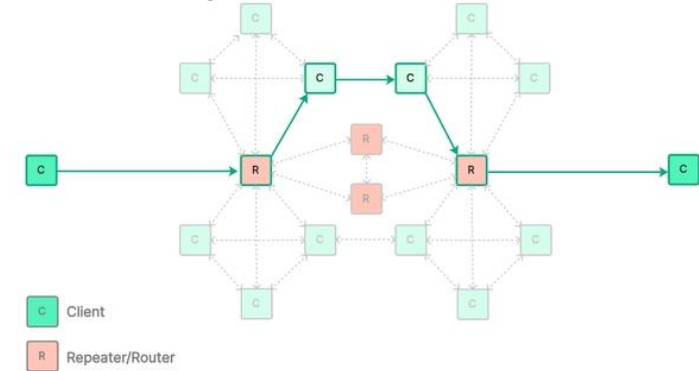
- Set of LoRa radios deployed all over Studley campus
- Monitored in real-time
- Currently establishing performance benchmarks for Meshtastic in true urban conditions
- Will also be used to compare routing protocol



# In the Future

- Collect more data!
- Conclude on:
  - How effective of LoRa meshes are for emergency communication
  - The best way to do so
- Examine the security of Meshtastic and MeshCore firmware

Meshtastic Routing



MeshCore Routing

